

Factor Returns: Boosting vs Deep Learning

Model choice for small-data financial tabular

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Abstract

A recurring practical question in quantitative finance is deceptively simple: on tabular financial data with only a few hundred to a few thousand observations, should a practitioner default to gradient boosting (GBM) or to a shallow deep-learning (DL) model? We narrow this question to a measured comparison. Using the public Fama-French 3-factor daily series (market excess return Mkt-RF, size SMB, value HML), we build lagged features and set up two prediction tasks, next-day market excess return as a regression target and its sign as a classification target, then compare a histogram-based gradient boosting model against a one-hidden-layer multilayer perceptron (MLP) under a strict time-ordered split. Varying the training window from 60 to 8,000 trading days and evaluating on the same fixed, most-recent 500-day window, we find that in the extreme low-data regime (60 training days) the MLP's out-of-sample R-squared collapses to -2.60, while the GBM degrades far more gracefully to -0.07. The gap narrows as training data grows, and by 4,000-8,000 training days the MLP's R2 (-0.09 to -0.08) modestly overtakes the GBM's (-0.13 to -0.10). Direction-prediction accuracy stays within 46.6-52.6% across all training sizes for both models, close to the majority-class baseline of 49.0%, and ROC-AUC mostly sits between 0.46 and 0.55. Two conclusions follow. First, neither GBM nor shallow DL finds a stable signal that beats market efficiency for next-day factor direction. Second, the two model families fail differently: GBM is the far safer default in the small-data regime, and that advantage narrows or reverses as data accumulates. This paper is methodology research, not investment advice, and does not claim profitability for any live trading strategy.

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1 Abstract

A recurring practical question in quantitative finance is deceptively simple: on tabular financial data with only a few hundred to a few thousand observations, should a practitioner default to gradient boosting (GBM) or to a shallow deep-learning (DL) model? We narrow this question to a measured comparison. Using the public Fama-French 3-factor daily series (market excess return Mkt-RF, size SMB, value HML), we build lagged features and set up two prediction tasks, next-day market excess return as a regression target and its sign as a classification target, then compare a histogram-based gradient boosting model against a one-hidden-layer multilayer perceptron (MLP) under a strict time-ordered split. Varying the training window from 60 to 8,000 trading days and evaluating on the same fixed, most-recent 500-day window, we find that in the extreme low-data regime (60 training days) the MLP’s out-of-sample R-squared collapses to -2.60, while the GBM degrades far more gracefully to -0.07. The gap narrows as training data grows, and by 4,000-8,000 training days the MLP’s R² (-0.09 to -0.08) modestly overtakes the GBM’s (-0.13 to -0.10). Direction-prediction accuracy stays within 46.6-52.6% across all training sizes for both models, close to the majority-class baseline of 49.0%, and ROC-AUC mostly sits between 0.46 and 0.55. Two conclusions follow. First, neither GBM nor shallow DL finds a stable signal that beats market efficiency for next-day factor direction. Second, the two model families fail differently: GBM is the far safer default in the small-data regime, and that advantage narrows or reverses as data accumulates. This paper is methodology research, not investment advice, and does not claim profitability for any live trading strategy.

2 Background: why model choice diverges on small tabular financial problems

Recent tabular machine-learning literature converges on a fairly consistent message: on structured row-and-column data, tree-based gradient boosting ensembles remain as strong as, or stronger than, deep learning’s more expressive architectures. This gap is reported to widen as sample counts shrink, feature scales become more heterogeneous, and feature interactions become more local and non-smooth. Financial factor data fits much of this profile. Daily observations look plentiful in raw count, but the effective amount of statistically independent information is much smaller once volatility clustering, autocorrelation, and regime shifts are accounted for, and once a problem is narrowed to a single strategy, sector, or emerging factor, the usable training sample often drops to a few hundred rows. Newly proposed factors, emerging-market universes, small-cap subsets, and sector-restricted strategies are typical examples.

The practical difficulty is that practitioners face this choice as an operational decision, not a theoretical one. Whether the impressive performance deep-learning papers report on large datasets transfers to a factor-prediction problem with 200-2,000 samples, or whether tree-based methods remain the safer default in that regime, is a question each practitioner has historically had to re-verify on their own data. This paper attempts one honest answer to that question, using a real financial factor time series rather than a generic tabular benchmark (credit scoring, housing prices), respecting the time-ordering constraint specific to financial data, and systematically varying training size.

In the same spirit, this paper does not attempt to answer “which model beats the market.” Instead it focuses on a narrower and more reproducible question one step earlier: on identical features, identical splits, and identical metrics, how does each model family’s out-of-sample performance move as training size changes? Even if no market-beating signal exists, this question remains valid and useful, because if the two model families fail differently, that failure pattern itself is a legitimate basis for model choice.

3 Method: GBM vs shallow DL, time-ordered splits, leakage control

3.1 Models

Both model families were deliberately left untuned, close to widely used defaults, and we state explicitly that no hyperparameter was adjusted to favor either side.

- **Gradient boosting (GBM):** scikit-learn’s `HistGradientBoostingRegressor` / `HistGradientBoostingClassifier`,

a histogram-based tree-splitting implementation in the same family as LightGBM. Fixed at 200 boosting iterations, max depth 4, learning rate 0.05.

- **Shallow deep learning (DL)**: scikit-learn’s `MLPRegressor` / `MLPClassifier` with one hidden layer of 32 units, ReLU activation, and L2 weight decay of $1e-3$. Early stopping on a validation split was enabled to let the model self-regularize somewhat against overfitting, with a maximum of 2,000 iterations.

Both models were trained on features standardized (z-scored) using the mean and standard deviation of the training window only; the same scaling was then applied, unmodified, to the test window (leakage control).

3.2 Time-ordered splits and leakage control

The most common methodological error on financial time series is random cross-validation. Shuffling observations before splitting lets future information leak into predictions of the past, which systematically inflates out-of-sample performance. Every split in this paper is strictly time-ordered. The most recent 500 trading days of the full sample are held out as a fixed evaluation window, and the window immediately preceding it is varied in size (60, 125, 250, 500, 1,000, 2,000, 4,000, 8,000 trading days) to repeat the experiment. Training and evaluation windows never overlap in any run, and every feature for a given date is constructed using only information dated strictly before that date (lagged features, Section 3.2).

This design is a simplified version of walk-forward validation. Rather than averaging over multiple overlapping evaluation windows, we use a single fixed, most-recent evaluation window. This was a deliberate simplification to keep compute manageable and the experiment reproducible; the corresponding trade-off is stated explicitly in Section 5.

3.3 Evaluation metrics

The regression task is scored with out-of-sample R-squared; the classification task with accuracy and ROC-AUC. For both tasks we also compute a trivial baseline using only training-window statistics (predict the training-window mean for regression, predict the training-window majority class for classification), reported in the same tables so readers can check whether either model actually beats the trivial baseline.

4 Data and experimental setup

4.1 Data source (real, public data)

The data used in this paper is **real, public data**, not synthetic. The underlying series is the Fama-French 3-factor daily time series (market excess return Mkt-RF, size premium SMB, value premium HML, risk-free rate RF) distributed by Professor Kenneth R. French’s data library. The network environment used to run this experiment could not reach Professor French’s original server directly, so we instead used the identical CSV file mirrored publicly on GitHub (the `vicmart11/Fama-French-3-Factor` repository). The file’s own header comment states it was “created by CMPT_ME_BEME_RETS_DAILY using the 202307 CRSP database,” which self-documents the origin and generation method of the underlying data. The downloaded file contains 25,543 daily observations from July 1, 1926 through July 31, 2023.

4.2 Feature construction (lagged, time-safe)

Since the raw data contains only three factors, we expanded the feature space using historical lags and rolling statistics rather than using the raw factors directly. Specifically, for each of Mkt-RF, SMB, and HML we constructed 1-, 2-, 3-, 5-, 10-, and 20-day lags (18 lag features), and added rolling means and rolling volatility of Mkt-RF plus rolling means of SMB and HML over 5-, 10-, and 20-day windows (4 features per window across 3 windows, 12 features), for 30 features in total. All rolling statistics were also shifted to use only information available as of one day prior before the rolling window was computed, so that same-day information could not leak into the features. The prediction targets are the same-day Mkt-RF value (regression) and its sign (classification, $up = 1$), and since every feature is constructed from information strictly before that date, this is structurally equivalent to a next-day prediction problem using only information available up to yesterday. Rows

with missing values from the lag/rolling computation (the earliest 20 trading days) were dropped, leaving a final feature sample of 25,523 rows.

4.3 Experimental setup summary

Item	Value
Data source	Real, Fama-French 3-factor daily, GitHub mirror
Observation period	1926-07-27 to 2023-07-31 (after feature construction)
Number of features	30 (18 lags + 12 rolling statistics)
Evaluation window	Most recent 500 trading days (fixed)
Training window sizes	60 / 125 / 250 / 500 / 1,000 / 2,000 / 4,000 / 8,000 trading days
GBM	HistGradientBoosting, 200 trees, depth 4, learning rate 0.05
DL	MLP, 1 hidden layer of 32 units, ReLU, L2=1e-3, early stopping
Seed	42 (fixed)

5 Results

5.1 Regression: GBM is far safer in the low-data regime

The table below reports out-of-sample R-squared by training size for GBM, MLP, and the trivial training-mean baseline. All three columns are measured, not simulated.

Train days	R2 (GBM)	R2 (MLP)	R2 (mean baseline)
60	-0.074	-2.600	-0.003
125	-0.295	-0.347	-0.005
250	-0.120	-0.205	-0.009
500	-0.156	-0.208	-0.006
1,000	-0.168	-0.219	-0.003
2,000	-0.167	-0.228	-0.002
4,000	-0.131	-0.089	-0.001
8,000	-0.096	-0.076	-0.001

The most striking result is at 60 training days. The MLP’s out-of-sample R2 is -2.60, meaning its predictions are far worse than simply predicting the training mean. Under the same condition the GBM degrades far more gracefully, to -0.07, a gap of nearly 35x. This is consistent with the general expectation that shallow neural networks overfit catastrophically with very few training samples, while tree ensembles’ split conditions are naturally more constrained and degrade less severely.

This gap narrows rapidly as training data grows. At 125 training days the difference between GBM (-0.295) and MLP (-0.347) is already within a single order of magnitude, and, notably, at 4,000 and 8,000 training days the MLP (-0.089, -0.076) modestly overtakes the GBM (-0.131, -0.096). It is important not to overread this: both models remain clearly worse than the mean baseline (near -0.001) in this regime as well. In other words, “MLP beats GBM” here means precisely that, among two models that both underperform a trivial baseline, the less-bad one switches to MLP, not that either model has achieved economically meaningful predictive power.

5.2 Classification: direction prediction is close to a coin flip for both models

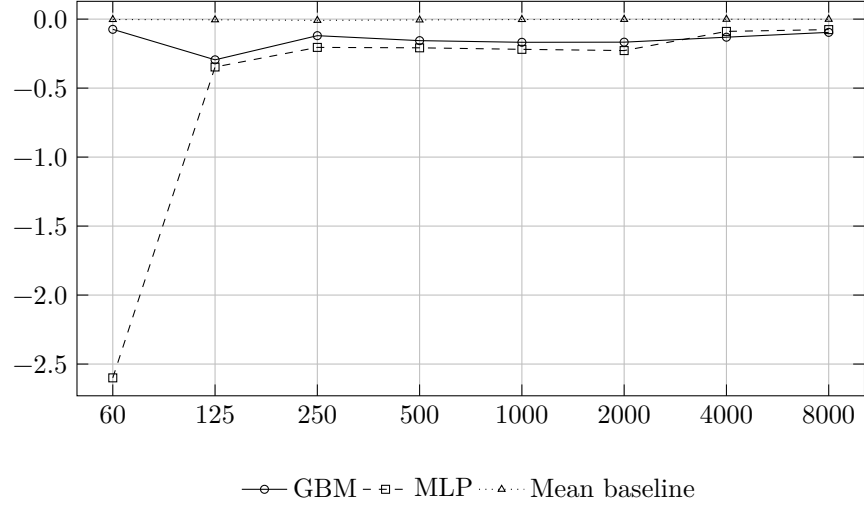


Figure 1: Out-of-sample R2 by training size

Train days	Accuracy (GBM)	Accuracy (MLP)	Majority baseline	AUC (GBM)	AUC (MLP)
60	0.470	0.496	0.490	0.480	0.483
125	0.502	0.490	0.490	0.507	0.483
250	0.526	0.490	0.490	0.546	0.475
500	0.516	0.492	0.490	0.531	0.485
1,000	0.500	0.488	0.490	0.504	0.485
2,000	0.468	0.502	0.490	0.468	0.504
4,000	0.466	0.524	0.490	0.462	0.521
8,000	0.484	0.500	0.490	0.511	0.518

Direction (up/down) prediction accuracy stays within 46.6%-52.6% across every training size and both models, indistinguishable in practice from the majority-class baseline (always predict up, 49.0%). ROC-AUC is also mostly between 0.46 and 0.55, not far from 0.5 (random). The highest AUC observed is the GBM at 250 training days (0.546), but this single value alone is not sufficient evidence of a stable edge, since it comes from a single evaluation window (see Section 5).

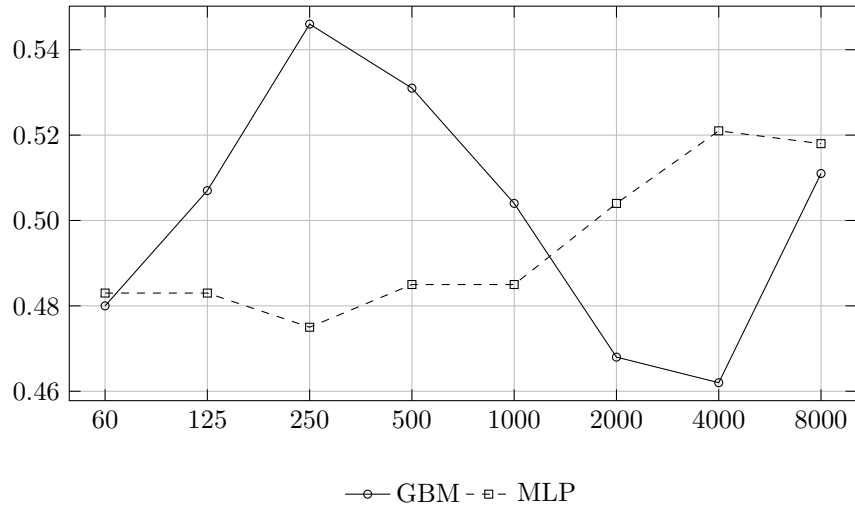


Figure 2: Direction-prediction ROC-AUC by training size

Nonetheless, the same pattern observed in the regression results appears here too. Between 250 and 1,000 training days, the GBM's AUC (0.50-0.55) consistently leads the MLP's (0.48-0.49), and beyond 2,000 training days the MLP's AUC (0.50-0.52) catches up to or overtakes the GBM's (0.46-0.51). The fact that the same crossover pattern appears independently in both the regression and classification tasks, in the same 2,000-4,000 training-day range, suggests this may reflect a real difference in how the two model families respond to data

strength, and learning-rate schedule) could change the results. The goal of this paper was not “a contest between the best-tuned version of each model” but observing “the tendency each model family shows under a reasonable default configuration.”

3. **Features are simple:** 30 features built purely from lags and rolling statistics are modest compared to the more sophisticated factor models used in practice (combining momentum, valuation, quality, and macroeconomic indicators). The relative advantage between the two models could differ under a richer feature set.
4. **Absolute level of the regression results:** R^2 being negative across every regime, and generally worse than the mean baseline, means this problem setup (predicting next-day market excess return from daily lagged factor features) found no meaningful predictive power beyond a trivial baseline. This is consistent with market efficiency, not an implementation defect in either model. However, we restate that the “GBM is better / MLP is better” comparisons in this paper are strictly comparisons of which model fails less, given that both largely fail.
5. **Not investment advice:** this is methodology research, not investment advice. It does not claim live profitability for any specific factor, strategy, or model. The data, code, and results in this paper are for research and reproducibility purposes only and should not be applied directly to live capital management.

7 Data and reproduction

- **Data:** Fama-French 3-factor daily time series (Mkt-RF, SMB, HML, RF). The original source is Professor Kenneth R. French’s data library (Tuck School of Business, Dartmouth College), which is widely published for academic research use. This experiment downloaded the identical CSV (`F-F_Research_Data_Factors_daily.CSV`) mirrored on the public GitHub repository `vicmarti1/Fama-French-3-Factor` via `raw.githubusercontent.com`, rather than from the original server. **Caveat:** this mirror repository carries no explicit license file of its own. The underlying data content is French’s publicly published academic data, so citation for research purposes is not a concern, but reproducers are encouraged to download directly from the original source (the Kenneth R. French Data Library) when possible. The mirror was used here only because the network environment running this experiment could not reach French’s original server (`dartmouth.edu`) directly; this is stated explicitly as an access limitation, not a reliability concern for the results.
- **Data nature:** entirely **real**. No synthetic data was used.
- **Seed:** fixed at 42, used consistently for standardization, GBM, and MLP.
- **Reproduction steps:** (1) download the CSV above, strip the header comment lines, and parse dates. (2) construct features from 1-, 2-, 3-, 5-, 10-, 20-day lags and 5-, 10-, 20-day rolling mean/volatility of Mkt-RF, SMB, HML. (3) hold out the most recent 500 trading days as the evaluation window, and use the preceding 60-8,000 trading days as the training window. (4) fit `HistGradientBoostingRegressor/Classifier` and `MLPRegressor/Classifier` (scikit-learn) on the training window and compute R^2 , accuracy, and ROC-AUC on the evaluation window. scikit-learn version 1.9, seed 42.

8 References

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